The Physics of the Run-24 p+p Pion (π^0/η) Nonlinearity Problem

What "Resolution" and "Smearing" Mean, and Why PYTHIA Has to be Smeared

An sPHENIX teaching note (EMCal calibration / PPG12 context)

2026-06-17

Abstract

This note teaches the physics behind a concrete sPHENIX problem: in Run-24 p+p collisions at $\sqrt{s} = 200\,\mathrm{GeV}$, the reconstructed π^0 and η mesons in the electromagnetic calorimeter (EMCal) do *not* look the same in data as in the PYTHIA-8 + GEANT4 simulation. The simulated mass peaks are **too narrow** (the MC "looks too good"), and the energy scale shows a residual **nonlinearity** versus transverse momentum. We explain, from first principles, (i) what *energy resolution* is in a calorimeter, (ii) what *smearing* of the Monte Carlo (MC) means and why it is needed, (iii) how the two connect through the standard three-term resolution formula, and (iv) why a perfectly good simulation must be deliberately "made worse" to do honest physics. All numerical parameters quoted are the ones used in the sPHENIX EMCal calibration talks and the PPG12 analysis note.

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1 The measurement and the problem

sPHENIX reconstructs π^0 and η mesons through their two-photon decay:

$$\pi^0 \to \gamma\gamma \quad (m_{\pi^0} = 0.134977\,\mathrm{GeV}), \qquad \eta \to \gamma\gamma \quad (m_\eta = 0.547862\,\mathrm{GeV}). \tag{1}$$

The two photons each deposit their energy as an electromagnetic *shower* in the EMCal, forming a localized cluster of fired towers. From the two cluster energies E_1, E_2 and their opening angle θ_{12} we form the **two-photon invariant mass**:

$$M_{\gamma\gamma} = \sqrt{2\, E_1 E_2\, (1 - \cos\theta_{12})}. \tag{2}$$

Plotting $M_{\gamma\gamma}$ for many photon pairs produces a peak sitting at the meson mass on top of a combinatorial background. Two numbers are read off the peak in bins of meson transverse momentum p_T:

- the **peak position** M_{fit}, which tests the EMCal *energy scale*. If we plot $M_{\text{fit}}/m_{\text{PDG}}$ versus p_T and it is *not flat*, the detector response is **nonlinear** in energy — this is the "nonlinearity" in the title.
- the **peak width** $\sigma(M)$, which tests the EMCal *energy resolution*. A broad peak means a blurry detector.

The Run-24 p+p symptom. When data are overlaid on the nominal PYTHIA+GEANT4 simulation, *the data peak is much wider than the simulated peak*, and $M_{\text{fit}}/m_{\text{PDG}}$ drifts with p_T by a couple of percent. The simulation is therefore both too sharp (resolution problem) and slightly mis-scaled (nonlinearity problem). The fix has two independent knobs: **smearing** to widen the MC peak, and an **energy-scale / nonlinearity correction** to re-center it.

Why does this matter? Because the final physics result — e.g. the PPG12 isolated-prompt-photon cross section, or a π^0/η spectrum — is obtained by *comparing data to a simulation*. Efficiencies, backgrounds, and the *unfolding* that corrects detector smearing back to truth are all computed from the MC. If the MC does not reproduce the true detector resolution and scale, every one of those corrections is biased. The simulation must tell the truth about how good (and how bad) the real detector is.

2 Energy resolution: what "blurry" means quantitatively

2.1 The definition

Send a photon of a *fixed* true energy E into the calorimeter many times. The reconstructed energies do not all come back equal; they scatter in a roughly Gaussian cloud of width σ_E around a mean. The **energy resolution** is the fractional width

$$\frac{\sigma_E}{E}. \tag{3}$$

A small σ_E/E means a sharp, well-measured detector; a large value means a blurry one. Crucially, the resolution is *not* a single number — it depends on energy, and that dependence carries physics.

2.2 The three-term calorimeter formula

For a sampling calorimeter like the sPHENIX EMCal (a tungsten–scintillating-fiber "SPACAL"), the resolution is parameterized by three terms added *in quadrature*:

$$\boxed{\frac{\sigma_E}{E} = \sqrt{\frac{p_0^2}{E} + \frac{p_1^2}{E^2} + p_2^2} \;\equiv\; \frac{p_0}{\sqrt{E}} \oplus \frac{p_1}{E} \oplus p_2} \qquad (E \text{ in GeV}), \tag{4}$$

where "$\oplus$" denotes addition in quadrature. Each term is a distinct piece of physics:

$p_0/\sqrt{E}$ — **stochastic (sampling) term.** Comes from *statistical fluctuations* in the shower: how many shower particles cross active scintillator, how many scintillation photons are made, how many photoelectrons the SiPMs detect. These are Poisson-like, so the relative fluctuation scales as $1/\sqrt{N} \propto 1/\sqrt{E}$. It **shrinks** as energy rises — high-energy showers are measured *proportionally* better.

p_1/E — **noise term.** Electronic noise and pileup add a roughly *fixed* amount of energy regardless of the signal, so as a *fraction* it falls like $1/E$. It matters only at low energy and is often negligible (here p_1 is small or zero).

p_2 — **constant term.** Comes from effects that scale *with* the energy: calibration spread tower-to-tower, light-collection non-uniformity, shower leakage out the back/sides, and dead material. Being a fixed fraction, it does **not** shrink with energy and therefore **dominates the resolution at high** E.

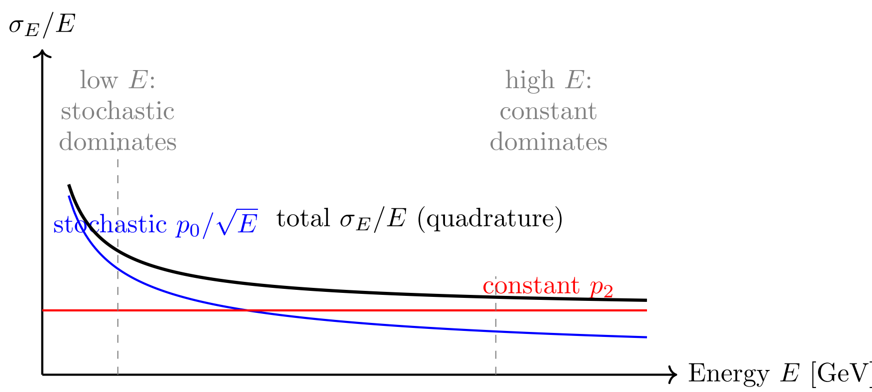

Figure 1. Schematic of the three-term resolution. The stochastic term (blue) falls as $1/\sqrt{E}$; the constant term (red) is flat; their quadrature sum (black) is the measured resolution. At high energy the constant term sets the floor — which is exactly where the Run-24 data/MC disagreement lives.

2.3 From photon resolution to meson width

Because Eq. (2) multiplies the two photon energies, the *meson* relative width inherits the photon resolution. For symmetric decays one finds, to good approximation,

$$\frac{\sigma(M)}{M} \approx \frac{1}{\sqrt{2}} \frac{\sigma_E}{E} \oplus \text{(angular / position term)}, \tag{5}$$

so the *width of the π^0/η peak versus p_T* is a direct, data-accessible measurement of the EMCal energy (and position) resolution. This is the experimental handle the calibration team uses: fit the peak in each p_T slice, plot $\sigma(M)/M$ vs p_T, and read off the resolution. The reference sPHENIX single-photon (test-beam) resolution is

$$\frac{\sigma_E}{E} \approx \frac{16\%}{\sqrt{E}} \oplus 5\%, \tag{6}$$

i.e. $p_0 \approx 0.16$, $p_2 \approx 0.05$.

3 Why PYTHIA needs "smearing"

3.1 What PYTHIA and the MC chain actually do

PYTHIA-8 (Detroit tune) generates the $p+p$ collision: it produces the partons, hadronizes them, and outputs a list of final-state particles with *exact, truth-level* four-momenta. PYTHIA itself knows nothing about the detector — it is pure event generation. Those truth particles are then passed to **GEANT4**, which simulates the showers in the EMCal, and finally the same reconstruction used on data turns the simulated energy deposits into clusters and meson candidates.

> **The core issue.** "PYTHIA smearing" is shorthand for the whole PYTHIA+GEANT4+reco simulation chain, and the empirical fact that *its reconstructed energy resolution is better (narrower) than the real detector's.* The simulated EMCal is intrinsically too clean: fitting the unsmeared MC photon response gives a resolution like $\sigma_E/E \approx 0.185/\sqrt{E} \oplus 0.040$ — note the very small constant term $p_2 \approx 0.04$. Real Run-24 data want a constant term closer to 0.05–0.08. The MC peak is therefore sharper than data, and no honest comparison is possible until the MC is broadened to match.

Why is the simulation too good? In practice, real detectors carry imperfections that are hard to capture fully in GEANT4: tower-to-tower miscalibration spread, light-collection non-uniformity along the fibers, SiPM-to-SiPM gain variation, subtle dead/edge material, and time-dependent effects in a real run. These inflate the *constant term* p_2 in data above what the idealized simulation produces. (The calibration talks note candidly that it is "unclear" exactly which effect dominates — which is precisely why the correction is applied empirically.)

3.2 What "smearing" is

Smearing is the deliberate, controlled *degradation* of a simulated quantity so that the MC reproduces the worse resolution of real data. It is the mirror image of resolution: resolution *measures* the blur; smearing *injects* the blur into an otherwise-too-perfect simulation.

Operationally, each simulated photon (or cluster) energy is multiplied by a random Gaussian factor centered on one:

$$E \longrightarrow E \cdot f, \qquad f \sim \mathrm{Gaus}\big(1,\ \sigma_{\mathrm{extra}}(E)\big). \tag{7}$$

The width σ_{extra} is *not* the full data resolution — the MC already has its own intrinsic resolution. We only need to add the *missing* amount, computed as the **quadrature difference** between the target (data) resolution and what the MC already has:

$$\boxed{\sigma_{\mathrm{extra}}(E) = \sqrt{\max\big(0,\ \sigma^2_{\mathrm{data}}(E) - \sigma^2_{\mathrm{MC}}(E)\big)}} \tag{8}$$

The $\max(0, \cdot)$ guards against the (unphysical) case where the MC is already wider than data — you can broaden a distribution but you cannot sharpen it by adding random noise. The position (angle) of the photon is smeared the same way, but *additively* on η and ϕ and centered at zero: $\eta \to \eta + \mathrm{Gaus}(0, \sigma_\psi)$, similarly for ϕ.

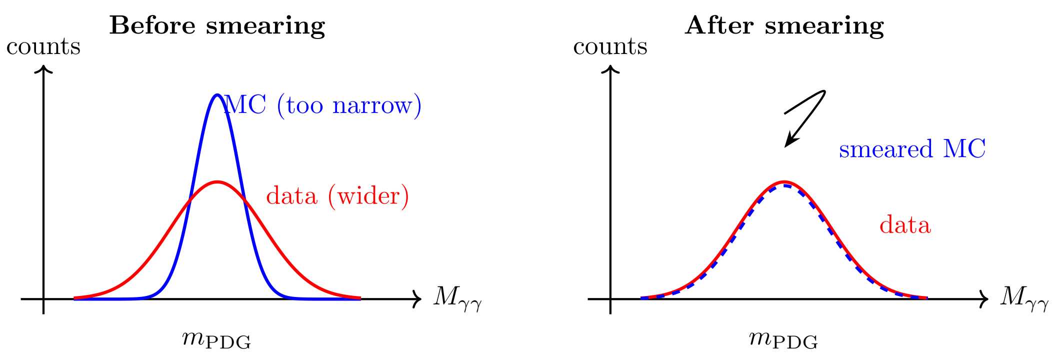

Figure 2. The role of smearing. Left: the nominal MC peak (blue) is sharper than the data peak (red) — this is the symptom. Right: after injecting σ_{extra} via Eq. (7), the MC width (blue dashed) is broadened to match data. Smearing changes the width, *not the* center*; re-centering is the separate energy-scale job.*

3.3 The smearing parameters actually used

The data target and MC resolutions are each fit to the three-term form Eq. (4). The numbers used in the sPHENIX calibration talks and the PPG12 analysis note are:

Resolution set	p_0 $(/\sqrt{E})$	p_1 $(/E)$	p_2 (const)
PYTHIA intrinsic / MC (too narrow)	0.185	0.00	0.040
nominal data target	0.15	0.05	0.05
"wider-data" variation (systematic)	0.13	0.08	0.08
tuned working set ("smear11")	0.180	0.020	0.015
test-beam single-γ reference	0.16	–	0.05

Reading the physics out of the table:

- The MC's constant term ($p_2 = 0.04$) is smaller than data's ($p_2 = 0.05$–0.08). That gap *at high energy* is the entire problem, because the constant term is the high-E floor (Fig. 1).
- With the nominal target, σ_{extra} from Eq. (8) is ≈ 0 below $\sim$17 GeV and rises to only $\sim$2% above 20 GeV. **Smearing barely matters at low p_T and matters most at high p_T** — consistent with the constant-term picture.
- The "wider-data" variation gives $\sigma_{\mathrm{extra}} \approx 6\%$ roughly flat in energy. The spread between "no extra smearing" and "wider-data" is taken as the *energy-resolution systematic* (the PPG12 "eres" uncertainty), reaching $\pm 9.6\%$ near $E_T \sim 27$ GeV.

The position smearing pairs with the above:

$$\sigma_\psi(E) = 0.0030/\sqrt{E} \oplus 0.0040 \quad \text{(tuned)}, \qquad \sigma_\psi(E) = 0.0055/\sqrt{E} \oplus 0.0020 \quad \text{(data-driven)}, \tag{9}$$

and a small energy-scale factor $E_{\mathrm{scale}} = f(E) \times 0.998$ re-centers the peak (the nonlinearity knob, Sec. 4).

3.4 The "smearN" samples

The simulation samples are named by their smearing recipe. The MC is fit, the quadrature-difference smear functions are stored, and applied photon-by-photon:

- pythia_smear1 — "global minimum difference" set.
- pythia_smear2 — extra constant $c_E = 0.08$; found *insufficient* (MC still too narrow).
- pythia_smear3 — extra constant $c_E = 0.05$.
- pythia_smear5 — "added an additional 8% energy smearing"; the nominal sample; width agreement at the MC statistical limit.
- pythia_smear11 — "added an additional $8\%/\sqrt{E}$" (i.e. an extra *stochastic* term, not a flat constant); the newest, leaving only a ∼2% high-p_T η residual.
- pythia_smear5_pix — smear5 plus the SiPM pixel-occupancy nonlinearity (Sec. 4).

The recurring "∼8% smearing" is just the extra ∼0.08 resolution term needed to widen the intrinsically narrow PYTHIA peak up to the data width.

4 The nonlinearity: re-centering the peak

Smearing fixes the *width*; a separate correction fixes the *mean*. A calorimeter is **nonlinear** if the reconstructed energy is not a constant fraction of the true energy — i.e. if $M_{\rm fit}/m_{\rm PDG}$ drifts with p_T. Two real, physical effects drive a mild nonlinearity in the sPHENIX EMCal, and both are added to the MC so it can describe data:

(1) Longitudinal light attenuation in the SPACAL fibers. A higher-energy shower develops *deeper* in the calorimeter (its maximum sits at depth $t_{\max} \sim X_0 \ln E$). Light produced deeper has a longer path to the SiPMs at the back, so more of it is attenuated:

$$\frac{S(E)}{E} \sim e^{-\Delta x/\Lambda} \sim E^{-X_0/\Lambda}, \qquad \Lambda \approx 1.6 \text{ m (TDR)}. \tag{10}$$

This costs roughly 1.0%/1.3%/1.6% of response at 10/20/40 GeV — a slow droop of the energy scale with energy. It is implemented as a z-dependent light efficiency in the MC.

(2) SiPM pixel saturation / occupancy. Each tower's SiPMs have a finite number of pixels, $N_{\rm pix} \sim 1.6 \times 10^5$ (4 SiPMs × ∼40k each), and fire ∼400–500 photoelectrons per GeV. As energy rises, an increasing fraction of pixels are already lit, so the response saturates. A first-order estimate gives $\sim 0.125\% \times E$, i.e. ∼1.25/2.5/5% at 10/20/40 GeV. This was missing from the nominal MC and is now added (validated against test-beam data) via pythia_smear5_pix.

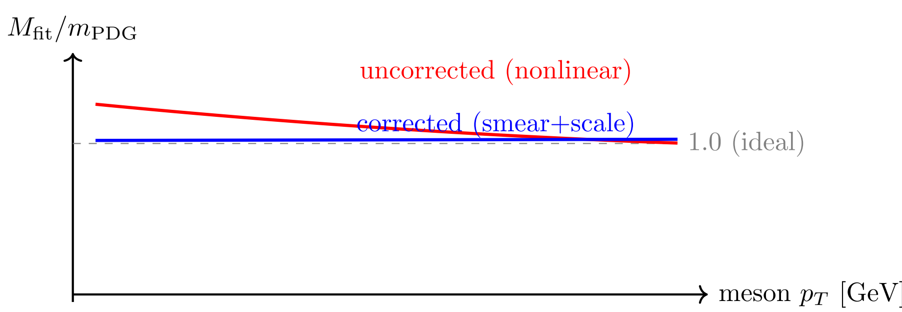

Figure 3. Schematic of the nonlinearity. Without the attenuation and pixel-saturation corrections, the reconstructed mass scale $M_{\rm fit}/m_{\rm PDG}$ drifts with p_T (red). With the MC scale correction $E_{\rm scale} = f(E) \times 0.998$ and the added physical effects, it is flat and close to unity (blue). This is a schematic, not measured data.

After tuning, the residual high-p_T η discrepancy is only about 2–2.5%, which is what the calibration update flags as the remaining "nonlinearity" to investigate.

5 Why this matters downstream: smearing feeds unfolding

A measured spectrum is the true spectrum *convolved* with detector effects (resolution, smearing, bin migration, efficiency). **Unfolding** is the statistical inversion that recovers truth from the measured distribution. With a *response matrix* $R_{ij} = P(\text{measured } i \,|\, \text{truth } j)$ built from MC,

$$\vec{m} = R\vec{t} \quad \Longrightarrow \quad \vec{t}_{\text{unfold}} = \mathcal{U}(R, \vec{m}), \tag{11}$$

solved with iterative Bayesian (D'Agostini) unfolding to avoid noise amplification. The response matrix R *is built from the MC*. If the MC resolution is wrong — too narrow — then R underestimates the true bin migration, and the unfolded result is biased. Toy studies in the sPHENIX work area deliberately mismatch the MC and data resolutions and show the unfolding bias grow as the mismatch grows. **This is the quantitative reason smearing must be tuned first**: an honest cross section (e.g. PPG12 isolated photons) requires a response matrix built from a simulation that has the same resolution as the data.

One-line glossary.
Resolution = how blurry the detector is (measured from the π^0/η peak width).
Smearing = injecting exactly the missing blur into the too-clean MC, via $E \to E \cdot \text{Gaus}(1, \sigma_{\text{extra}})$ with $\sigma_{\text{extra}} = \sqrt{\sigma^2_{\text{data}} - \sigma^2_{\text{MC}}}$.
Nonlinearity = the energy scale drifting with p_T, fixed by a separate E_{scale} correction plus added physical effects (fiber attenuation, SiPM saturation).
Unfolding = removing the detector blur from a measured spectrum to recover truth — which only works if the MC used to build it was smeared to match data.

6 Summary

1. In Run-24 p+p, the reconstructed π^0/η peaks are *wider in data than in PYTHIA+GEANT4* (resolution mismatch) and the mass scale drifts slightly with p_T (nonlinearity).
2. Resolution follows $\sigma_E/E = p_0/\sqrt{E} \oplus p_1/E \oplus p_2$; the data/MC gap lives in the *constant term* p_2, so it grows at high p_T.
3. "PYTHIA smearing" means broadening the too-clean simulated energy by a Gaussian factor of width $\sigma_{\text{extra}} = \sqrt{\sigma^2_{\text{data}} - \sigma^2_{\text{MC}}}$, with the working set $\sigma_E/E \approx 0.18/\sqrt{E} \oplus 0.02/E \oplus 0.015$ (sample "smear11", the recurring "$\sim$8%" extra term).
4. The nonlinearity is re-centered with $E_{\text{scale}} = f(E) \times 0.998$ plus two physical effects in the MC: longitudinal fiber light attenuation and SiPM pixel saturation.
5. All of this is required because the MC builds the efficiency, background, and *unfolding* corrections for the final cross section; a simulation that does not reproduce the real resolution would bias the physics result.

Numerical parameters are taken from the sPHENIX EMCal calibration update talks (B. Seidlitz) and the PPG12 analysis note (sPH-ppg-2024-012, v4). Figures here are hand-drawn TikZ schematics intended to illustrate the concepts; they are not measured data plots.